Experiment Design: Analysis of Muon with Nesterov Momentum

1 Overview

This document describes the experiment plan for the paper "Analysis of Muon with Nesterov Momentum." First implement all candidate methods, then use ablations to choose the randomized configuration, and finally compare the chosen method against strong baselines.

Exp.	**Purpose**	**Output**
E1	Ablate inexact solvers	solver selected
E2	Sweep normalized rank k/d	Selected rank
E3	Sweep NS steps q	Selected q
E4	Sweep batch size B	Selected batch size
E5	Compare the final randomized method against baselines	Final table and training curves
E6 (optional)	Explore sample complexity via ε-threshold stopping	Empirical K versus ε trend

Execution order: implement all candidate methods → E1 → E2 → E3 → E4 → E5. Run E6 only after the main comparison is stable.

Shared defaults before selection. Unless an experiment is actively sweeping a hyperparameter, keep the randomized Muon setup fixed at Nesterov momentum, oversampling $p = 10$, one power iteration $h = 1$, provisional NS depth $q = 5$, provisional normalized rank $k = d/8$, and provisional CIFAR-10 batch size $B = 2000$ and Nanogpt batch size $B = 1024$. The provisional values for k, q, and B are only working defaults for earlier ablations; later experiments replace them with the selected values.

Evaluation metric definition. We use Validation Perplexity for defining the accuracy for Nanogpt and accuracy for the CIFAR-10.

2 Experiment E1: Inexact Solver Ablation

2.1 Goal

Compare candidate inexact solvers and choose the solver that gives the best accuracy.

2.2 Protocol

1. Fix a shared model, training budget, data split, oversampling $p = 10$, power iteration $h = 1$, provisional rank $k = d/8$, default NS depth $q = 5$, and batch size $B = 2000$ for CIFAR-10 or $B = 1024$ for Nanogpt.
2. Keep Nesterov momentum fixed so that the only solver-side change is the inexact orthogonalization method.
3. Compare four inexact solvers:
   (a) Cubic Newton-Schulz: $f(x) = \frac{3}{2}x - \frac{1}{2}x^3$
   (b) Quintic Newton-Schulz with theoretical coefficients: $f(x) = 2x - \frac{3}{2}x^3 + \frac{1}{2}x^5$
   (c) Quintic Newton-Schulz with empirical coefficients: $(a, b, c) = (3.4445,\ -4.7750,\ 2.0315)$
   (d) Polar Express (degree-5)
4. For each solver, record training loss, validation/test accuracy, runtime.

2.3 Table and Figure

- one validation loss-versus-iteration figure with one line per solver.
- one final accuracy table with one row per solver.

3 Experiment E2: Rank Sweep

3.1 Goal

Choose the rank k that gives the best performance once the solver has been fixed.

3.2 Protocol

1. Fix the solver selected in E1 and keep Nesterov momentum, oversampling $p = 10$, power iteration $h = 1$, default NS depth $q = 5$, and batch size $B = 2000$ for CIFAR-10 or $B = 1024$ for Nanogpt unchanged.
2. Sweep ranks $k \in \{d/16,\ d/8,\ d/4,\ d/2\}$.
3. For each rank, record training loss, validation/test accuracy, runtime.

3.3 Outputs

- one accuracy-versus-iteration figure with one line per rank.
- one final accuracy table with one row per rank.
- one singular value decay plot with one line per rank on the last convolution layer of the ConvGroup in CIFAR-10 or the Q,K,V matrices in Nanogpt. (Similar to Figure 5 in the paper: "Low-rank orthogonalization for large-scale matrix optimization with applications to foundation model training".) This is to check the momentum matrix is approximated well enough by the rank-k approximation. We want to say the singular value decay fast.

4 Experiment E3: NS-Step Sweep

4.1 Goal

Choose the NS depth q that gives the best performance after the solver and rank have been fixed.

4.2 Protocol

1. Fix the solver selected in E1, the rank selected in E2, Nesterov momentum, oversampling $p = 10$, power iteration $h = 1$, and batch size $B = 2000$.
2. Sweep $q \in \{1, 3, 5, 7, 9\}$.
3. For each q, record training loss, validation/test accuracy, runtime.

4.3 Outputs

- one accuracy-versus-iteration figure with one line per NS depth.
- one final accuracy table with one row per NS depth.

5 Experiment E4: Batch-Size Sweep

5.1 Goal

Choose a batch size B after the solver, rank, and NS depth have been fixed.

5.2 Protocol

1. Fix the solver selected in E1, the rank selected in E2, the NS depth selected in E3, Nesterov momentum, oversampling $p = 10$, and power iteration $h = 1$.
2. Sweep batch sizes $B \in \{500,\ 1000,\ 2000, 3000, 4000\}$ for CIFAR-10 and $B \in \{256,\ 512,\ 1024, 2048, 4096\}$ for Nanogpt.
3. For each batch size, record end-to-end accuracy, runtime, throughput, and stability.

5.3 Outputs

- one accuracy-versus-iteration figure with one line per batch size.
- one final accuracy table with one row per batch size.

6 Experiment E5: Comparison with External Methods

6.1 Configuration of Our Method

Before running E5, determine all hyperparameters from the preceding ablations:

Parameter	**Determined by**
Orthogonalization solver	E1
Normalized rank k/d	E2
q (NS steps)	E3
Batch size B	E4
Oversampling p	Fixed ($p = 10$)
Power iteration h	Fixed ($h = 1$)
β (momentum)	Bayesian search
Momentum type	Fixed (Nesterov)
Learning rate η	Bayesian search

6.2 Table and Figure

Summary table:

Method	**Accuracy (mean ± std)**	**Time (sec)**
SGD + Nesterov		
AdamW		
Muon (Polyak, full NS)		
Muon (Nesterov, full NS)		
Muon (Polyak, randomized NS)		
Muon (Nesterov, randomized NS)		

Figure:
- one accuracy-versus-iteration figure, one line per method.
- one final accuracy table with one row per method.

7 Experiment E6: Sample Complexity

7.1 Goal

Explore the empirical relation between a target accuracy threshold ε and the number of steps needed to reach it.

7.2 Protocol

1. Fix oversampling $p = 10$, power iteration $h = 1$, the solver selected in E1, the rank selected in E2, the NS depth selected in E3, and the batch size selected in E4. But we need to preselect the max number of iterations K_{max} for choosing η to match the theory such as $\eta = K_{max}^{-1/2}$.
2. Choose thresholds $\varepsilon \in \{1e-1, 1e-2, 1e-3, 1e-4, 1e-5\}$.
3. For each ε, train until the monitored gradient norm first satisfies $\|g_k\| \leq \varepsilon$. We also want to run more trails for the same ε to get a more stable result. And then use the mean of the stopping steps to plot.
4. Record the stopping step K or K_{max} and compute the mean of the stopping steps.
5. Plot the mean of the stopping steps versus ε and check whether the empirical trend is consistent with the order suggested in A.4 Sample Complexity (Inexact Case) .

7.3 Output

- one plot of stopping step K versus ε;

8 CIFAR-10 Results

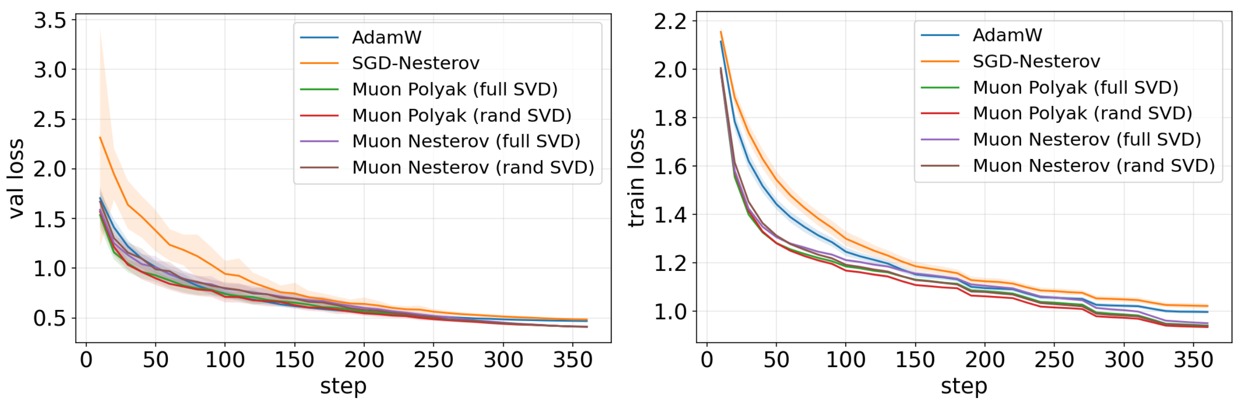
All curves are averaged over 50 trials; shaded bands show ±1 standard deviation. Test accuracies are reported as mean ± std over trials.

Fixed parameters across experiments. Each ablation sweeps one parameter while fixing the others; later experiments use the values selected by earlier ones.

Exp.	Solver	Rank k	NS iterations q	Batch size B
E1	sweep	256	5	2000
E2	Quintic (theoretical)	sweep	5	2000
E3	Quintic (theoretical)	256	sweep	2000
E4	Quintic (theoretical)	256	9	sweep
E5	Quintic (theoretical)	256	9	1000

Table 1: Configuration used in each experiment. "sweep" marks the parameter varied in that row; "prov." marks a provisional value used before the corresponding ablation has been run.

8.1 E5: Comparison with External Methods



(a) Validation loss

(b) Training loss

Figure 1: E5: validation and training loss for each method.

Method	Test accuracy (mean $\pm$ std)
AdamW	0.9106 ± 0.0026
SGD-Nesterov	0.9091 ± 0.0035
Muon Polyak (full SVD)	0.9380 ± 0.0013
Muon Polyak (rand SVD)	0.9378 ± 0.0012
Muon Nesterov (full SVD)	0.9377 ± 0.0011
Muon Nesterov (rand SVD)	$\mathbf{0.9383 \pm 0.0014}$

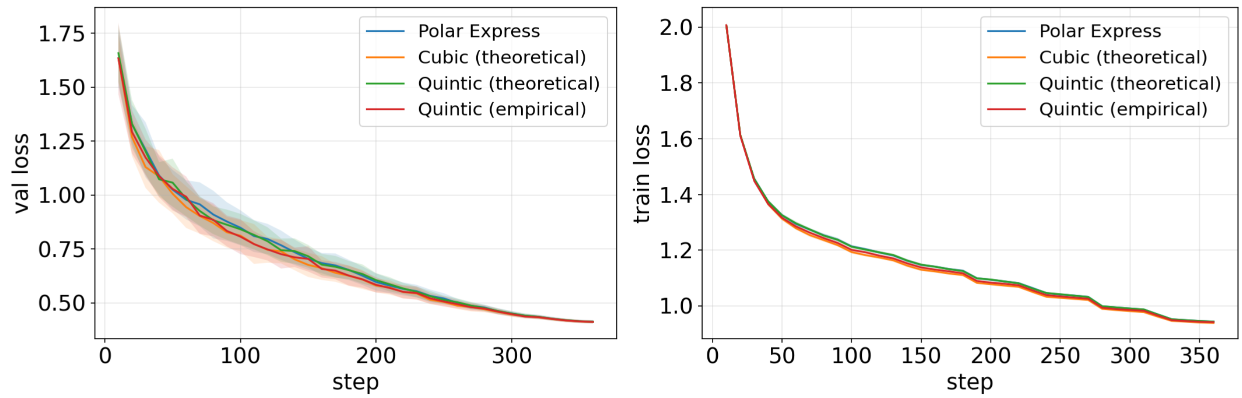
Table 2: E5: final TTA test accuracy.

8.2 E1: Solver Ablation

8.3 E2: Rank Sweep

8.4 E3: NS-Step Sweep

8.5 E4: Batch-Size Sweep

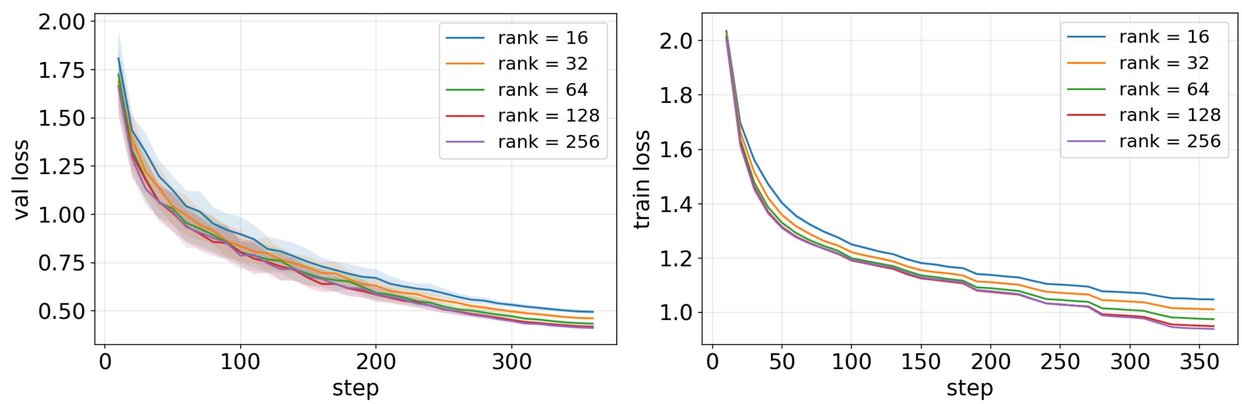


(a) Validation loss (b) Training loss

Figure 2: E1: validation and training loss by solver.

Solver	Test accuracy (mean $\pm$ std)
Polar Express	0.9384 ± 0.0014
Cubic (theoretical)	0.9383 ± 0.0012
Quintic (theoretical)	$\mathbf{0.9384 \pm 0.0016}$
Quintic (empirical)	0.9382 ± 0.0013

Table 3: E1: final TTA test accuracy by solver.

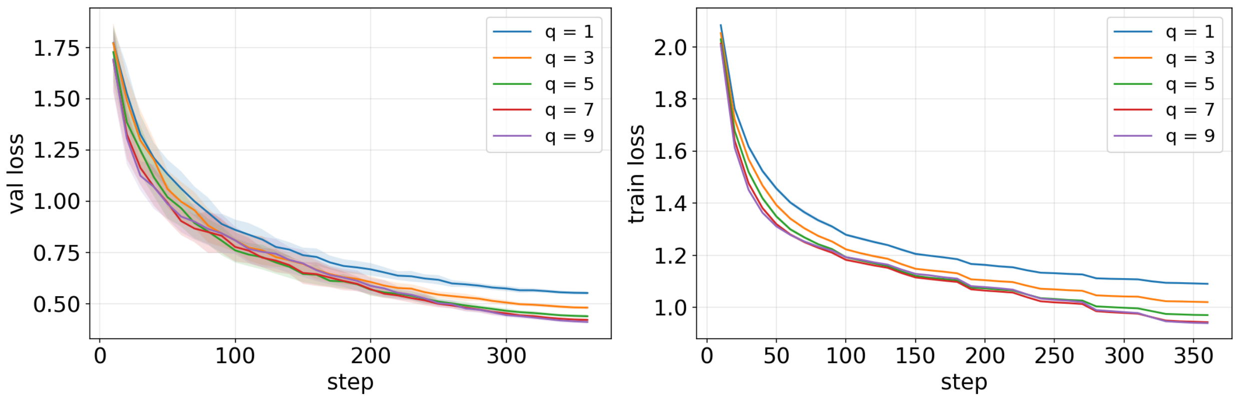


(a) Validation loss (b) Training loss

Figure 3: E2: validation and training loss by rank.

Rank	Test accuracy (mean $\pm$ std)
rank = 16	0.9068 ± 0.0014
rank = 32	0.9180 ± 0.0014
rank = 64	0.9284 ± 0.0015
rank = 128	0.9357 ± 0.0014
rank = 256	$\mathbf{0.9381 \pm 0.0014}$

Table 4: E2: final TTA test accuracy by rank.

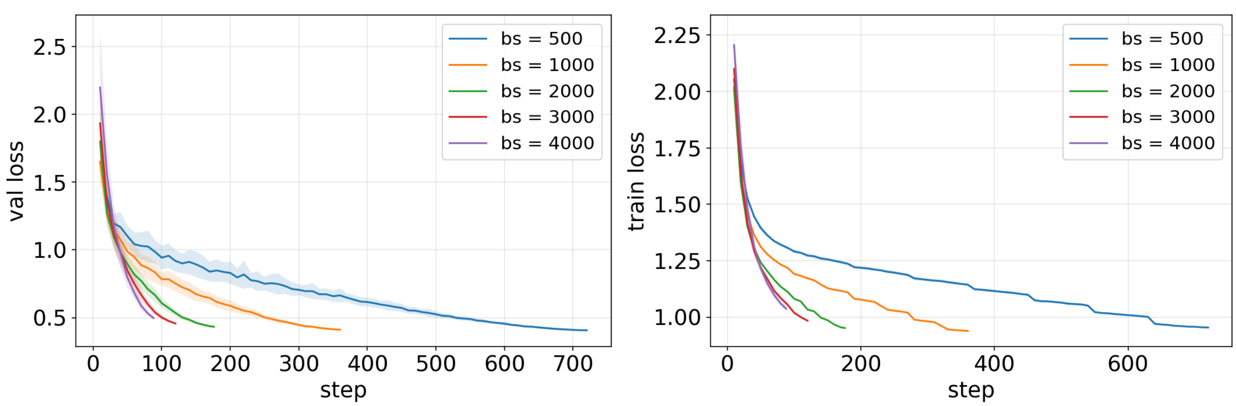


(a) Validation loss (b) Training loss

Figure 4: E3: validation and training loss by NS iterations.

NS iterations	**Test accuracy (mean ± std)**
$q = 1$	0.8940 ± 0.0019
$q = 3$	0.9165 ± 0.0013
$q = 5$	0.9296 ± 0.0016
$q = 7$	0.9362 ± 0.0012
$\mathbf{q = 9}$	$\mathbf{0.9386 \pm 0.0015}$

Table 5: E3: final TTA test accuracy by NS iterations.



(a) Validation loss (b) Training loss

Figure 5: E4: validation and training loss by batch size.

Batch size	**Test accuracy (mean ± std)**
$B = 500$	0.9373 ± 0.0013
$\mathbf{B = 1000}$	$\mathbf{0.9386 \pm 0.0016}$
$B = 2000$	0.9339 ± 0.0015
$B = 3000$	0.9272 ± 0.0022
$B = 4000$	0.9120 ± 0.0047

Table 6: E4: final TTA test accuracy by batch size.